1. Pseudocode for Detecting and Handling Outliers

Algorithm: Outlier Detection and Handling using Z-Score/IQR

Input: Dataset `D`
Output: Cleaned dataset `D`

BEGIN

1. Load the dataset D

2. Identify all numerical columns

3. FOR each numerical column C in D:

    a. Remove or handle missing values in C

    b. Choose an outlier detection method:

```
OPTION 1: Z-SCORE
Calculate:
   Mean = average(C)
   Standard Deviation = std(C)
   Z = (C - Mean) / Standard Deviation

Mark values where:
   |Z| > 3
as outliers

OPTION 2: IQR
Calculate:
   Q1 = 25th percentile of C
   Q3 = 75th percentile of C
   IQR = Q3 - Q1

Calculate:
   Lower Limit = Q1 - 1.5 × IQR
   Upper Limit = Q3 + 1.5 × IQR
```

Mark values outside these limits
as outliers

c. Count the detected outliers

d. Check whether the outlier is:
- Data entry error
- Invalid value
- Genuine extreme value

e. IF outlier is an invalid/error value:
Remove the record
ELSE IF outlier is genuine but extreme:
Cap the value at the lower/upper limit
ELSE:
Keep the value

4. Recalculate important statistics
such as Mean, Median, Minimum and Maximum

5. Validate the cleaned dataset:
- Check for remaining extreme values
- Check for missing values
- Check data types
- Check minimum and maximum values

6. Compare the dataset before and after cleaning

7. Save the cleaned dataset

8. Return cleaned dataset D

END

Simple Flow

Load Dataset
↓
Select Numerical Columns
↓
Calculate Z-Score / IQR
↓
Identify Outliers
↓
Check Whether Outlier Is Valid

↓
Remove / Cap / Keep
↓
Validate Dataset
↓
Save Clean Dataset

Result: The algorithm removes incorrect extreme values while preserving genuine observations, making the dataset more reliable for further analysis.

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2. Pseudocode for Data Aggregation and Summarization

Algorithm: Grouping and Aggregating Data

Input: Dataset D, grouping attributes, aggregation functions
Output: Summarized report

BEGIN

1. Load dataset D

2. Identify:
    - Grouping attributes
    - Numerical columns
    - Required metrics

3. Check for missing and invalid values

4. Clean the required columns

5. Sort the dataset based on the
   selected grouping attributes

6. Create an index for efficient access
   to grouped records

7. Group the dataset according to
   the specified attributes

8. FOR each group:

    a. Calculate SUM
      Total = SUM(values)

    b. Calculate AVERAGE
      Average = SUM(values) / COUNT(values)

    c. Calculate COUNT
      Count = number of records

    d. Store the calculated results

9. Combine all group results
  into a summarized table

10. Sort the summarized results
   according to the required metric

11. Validate the aggregation:

    a. Check total count:
      Sum of group counts
      = Total number of valid records

    b. Check total sum:
      Sum of group totals
      = Overall dataset total

    c. Check averages:
      Verify each average is within
      the minimum and maximum values
      of its group

12. If validation fails:
    Identify and correct the error

13. Generate the final summarized report

14. Display or save the report

END

Example

Suppose the dataset contains:

Category	Sales
Electronics	1000
Electronics	2000
Clothing	500
Clothing	1500

After grouping by **Category**:

Category	Total Sales	Average Sales	Count
Clothing	2000	1000	2
Electronics	3000	1500	2

Simple Flow

Load Dataset
↓
Select Grouping Attributes
↓
Clean and Validate Data
↓
Sort Data
↓
Create Index
↓
Group Records
↓
Calculate SUM / AVERAGE / COUNT
↓
Create Summary Table
↓
Sort Results
↓
Validate Results
↓
Generate Final Report